

Midterm 3

Name: _____

Student ID: _____

By signing below, you affirm that you have neither given nor received unauthorized help on this exam.

Signature: _____

DO NOT OPEN THE EXAM UNTIL YOU ARE TOLD TO DO SO! YOU WILL GET A ZERO IF YOU DO.

This exam contains 4 pages (including this cover page) and 4 problems. The exam will be graded out of 50 points. **This is an individual exam.** You are not allowed to use a calculator, any additional notes, look at anyone else's paper, have extra paper, talk to anyone, consult the internet, anything. It is considered cheating if you use additional notes, have wandering eyes, are caught talking, etc. Absolutely no headphones, smartwatches, or cellphones out! **I have a zero tolerance policy for cheating.**

Instructions: Print your name or student ID number above AND ON EVERY PAGE. You may use any available space on the exam for scratch work.

Good Luck!

To Receive Full Credit

- **Answer must be written in the box provided!** Show your work on the rest of the page.
- You must **show your work** to receive credit.
- **Proper Notation and Organized.** Clear, concise, and correct use of mathematical notation is part of a correct answer. Work should be organized in a reasonable neat and coherent way in the space provided. Work scattered all over the page without a clearing ordering is not acceptable.

Problem	Points	Score
1	10	
2	15	
3	15	
4	10	
Total:	50	

Do not write in the table to the right.

1. (10 points) Consider the IVP

$$y' = t + ty, \quad y(0) = 1$$

Starting with $y_0(t) \equiv 1$, find the Picard iterates $y_1(t), y_2(t), y_3(t), y_4(t)$. Box each Picard iterate.

$$y_0(t) = 1$$

$$y_1(t) = 1 + \int_0^t f(s, y_0(s)) ds$$

$$= 1 + \int_0^t s + s(1) ds = 1 + \left[\frac{s^2}{2} \Big|_0^t \right] = \boxed{1 + \frac{1}{2}t^2}$$

$$y_2(t) = 1 + \int_0^t s + s(1 + s^2) ds = 1 + \left[\frac{s^2}{2} + \frac{1}{4}s^4 \Big|_0^t \right] = \boxed{1 + \frac{1}{2}t^2 + \frac{1}{4}t^4}$$

$$y_3(t) = 1 + \int_0^t s + s(1 + s^2 + \frac{1}{4}s^4) ds$$

$$= 1 + \left[\frac{s^2}{2} + \frac{1}{4}s^4 + \frac{1}{24}s^6 \Big|_0^t \right] = \boxed{1 + \frac{1}{2}t^2 + \frac{1}{4}t^4 + \frac{1}{24}t^6}$$

$$y_4(t) = 1 + \int_0^t s + s(1 + s^2 + \frac{1}{4}s^4 + \frac{1}{24}s^6) ds$$

$$= 1 + \left[\frac{s^2}{2} + \frac{1}{4}s^4 + \frac{1}{48}s^6 + \frac{1}{24(8)}s^8 \Big|_0^t \right]$$

$$= \boxed{1 + \frac{1}{2}t^2 + \frac{1}{4}t^4 + \frac{1}{24}t^6 + \frac{1}{24(8)}t^8}$$

2. Consider the IVP $y' = -3y + \sin t$, $y(0) = 1$.

- (a) (10 points) Determine either a Lipschitz constant L or an interval on which $f(t, y)$ is Lipschitz in y . Justify your answer.

$$\begin{aligned} |f(t, y_1) - f(t, y_2)| &= |-3y_1 + \sin t + 3y_2 - \sin t| \\ &= |-3(y_1 - y_2)| \\ &\leq 3|y_1 - y_2| \end{aligned}$$

$$L = 3$$

Lipschitz w/ Lipschitz constant 3
for all real numbers

- (b) (5 points) State which part of the Picard-Lindelof Theorem is most applicable to this IVP, explain what it means in regards to region of the existence and uniqueness of a solution.

We can use part 1.

Since $f: [-c, c] \times \mathbb{R}$ and f is continuous & Lipschitz in y on $[-c, c] \times \mathbb{R}$. By P-L, we can conclude that there is exactly one sol'n $y: [-c, c] \rightarrow \mathbb{R}$ for any choice of interval $[-c, c]$, $c > 0$.

3. Consider the IVP $y' = \frac{y^2 + t}{t - 3}$, $y(0) = 1$

- (a) (10 points) Use Picard-Lidélöf (or its corollary) to show that there is $\delta > 0$ such that the IVP has a unique solution $y: [-\delta, \delta] \rightarrow \mathbb{R}$. Justify your answer.

Note $t \neq 3$

$$f(t, y) = \frac{y^2 + t}{t - 3}$$

$$\begin{aligned} |f(t, y_1) - f(t, y_2)| &= \left| \frac{y_1^2 + t}{t - 3} - \frac{y_2^2 + t}{t - 3} \right| = \left| \frac{y_1^2 - y_2^2}{t - 3} \right| \\ &= \left| \frac{1}{t - 3} \right| |y_1^2 - y_2^2| \end{aligned}$$

just needs to be finite, not include $t=3$ and contain the pt (0,1)

$$\leq \left| \frac{y_1 + y_2}{t - 3} \right| |y_1 - y_2|$$

↑ we need finite interval for y

Let $f: [-1, 1] \times [0, 2] \rightarrow \mathbb{R}$, now f has bounded derivative on $[0, 2]$ wrt y and so f is Lipschitz in y on $[-1, 1] \times [0, 2]$. Then P-L 2 says that there is a $\delta > 0$ s.t. we have a unique sol'n $y: (-\delta, \delta) \rightarrow \mathbb{R}$.

- (b) (5 points) Discuss the possible value of the $\delta > 0$. (Hint: $\delta = \min \left\{ \frac{1}{L}, \frac{b}{M} \right\}$)

$$\left| \frac{y^2 + t}{t - 3} \right| \leq \frac{4 + 1}{2} = \frac{5}{2} = M$$

$$[0, 2] = [-1, 1 + 1] \Rightarrow b = 1$$

$$\left| \frac{y_1 + y_2}{t - 3} \right| \leq \frac{4}{2} = 2 = L$$

$$\delta = \min \left\{ \frac{1}{2}, \frac{2}{5} \right\}$$

$\delta = \frac{2}{5}$ depends on initial y interval

4. (10 points) Consider the differential equation $y' = e^{ty}(y-3)$. For what initial conditions (t_0, y_0) are we guaranteed a unique solution near t_0 ? Justify your answer.

$$f(t, y) = e^{ty}(y-3)$$

$f(t, y)$ is continuous, bounded and has a bounded derivative with respect to y .

The initial condition can go anywhere and have a unique sol'n near t_0

